# Virtual Audio Router

VST3 Plugin Suite — User Manual

Version 2.1.0

# 1. Product Overview

**Virtual Audio Router** is a Windows virtual audio driver + VST3 plugin suite that enables zero-latency audio routing between:

- **System audio** and your **DAW** (Digital Audio Workstation)
- **DAW** audio to your **phone** for mobile monitoring
- **DAW** audio between multiple **PCs** on the same LAN

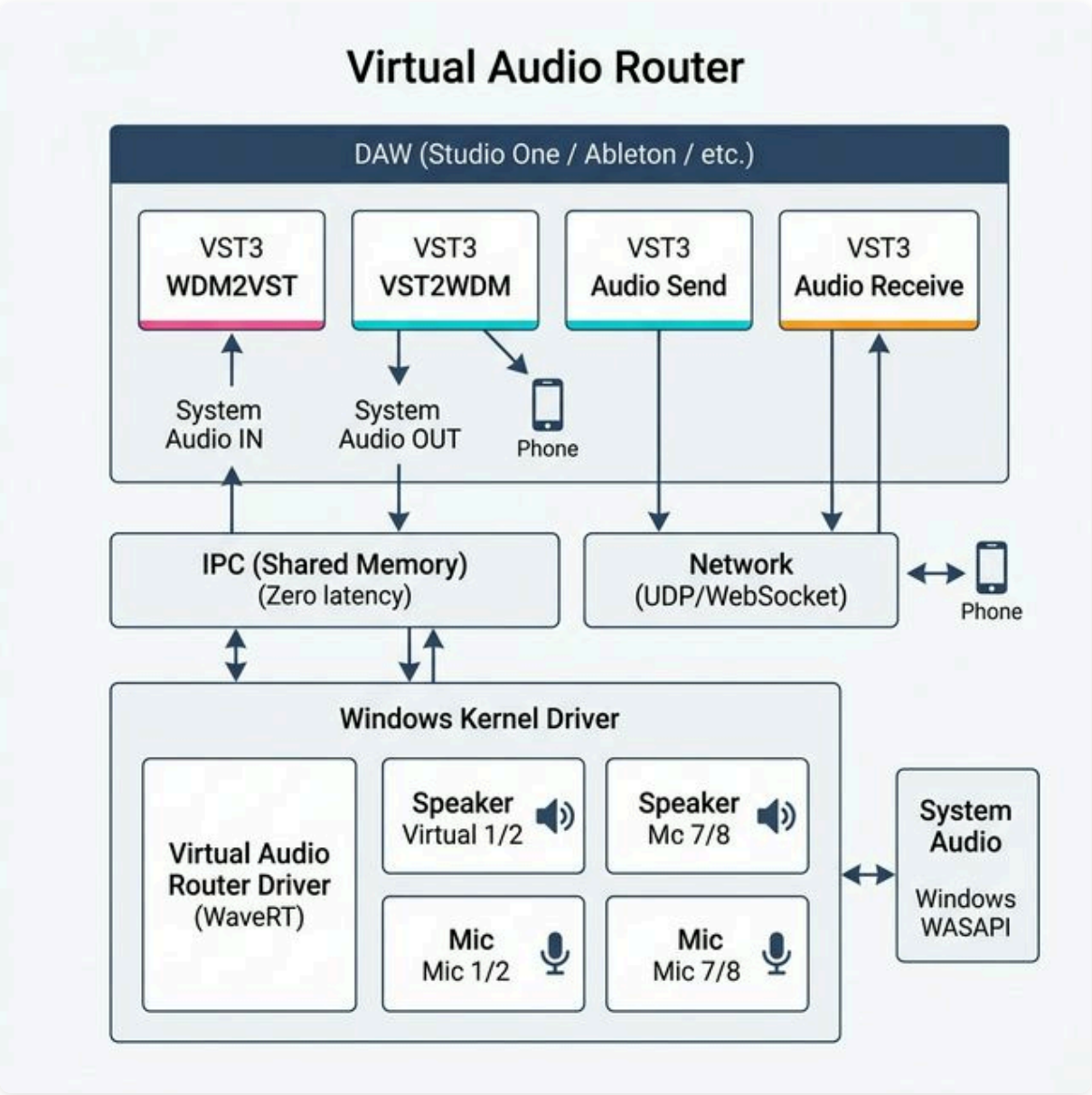


Figure 1: System Architecture Overview

## Components

| Component | Type | Description |
|---|---|---|
| Virtual Audio Router Driver | Windows Kernel Driver (WaveRT) | Creates virtual Speaker & Mic endpoints |
| WDM2VST | VST3 Plugin | Captures system audio into DAW |
| VST2WDM | VST3 Plugin | Sends DAW audio to system output + mobile |
| Audio Send | VST3 Plugin | Streams DAW audio over LAN / to mobile |
| Audio Receive | VST3 Plugin | Receives LAN audio into DAW |

# 2. System Requirements

| Requirement | Specification |
|---|---|
| OS | Windows 10/11 (64-bit) |
| DAW | Any VST3 host (Studio One, Ableton, Reaper, FL Studio, etc.) |
| CPU | x64 processor |
| RAM | 4 GB+ recommended |
| Network | LAN for Send/Receive; WiFi for mobile monitoring |

# 3. Installation

## 3.1 Driver

1. Run `VirtualAudioRouter_Setup.exe` as **Administrator**
2. Accept the driver signing prompt
3. Restart your computer
4. Verify in **Sound Settings**: Virtual 1/2 ~ 7/8 (playback) and Mic 1/2 ~ 7/8 (recording)

## 3.2 Plugins

Copy `.vst3` folders to:

```
C:\Program Files\Common Files\VST3\
```

Then rescan plugins in your DAW.

# 4. Plugin Usage Guide

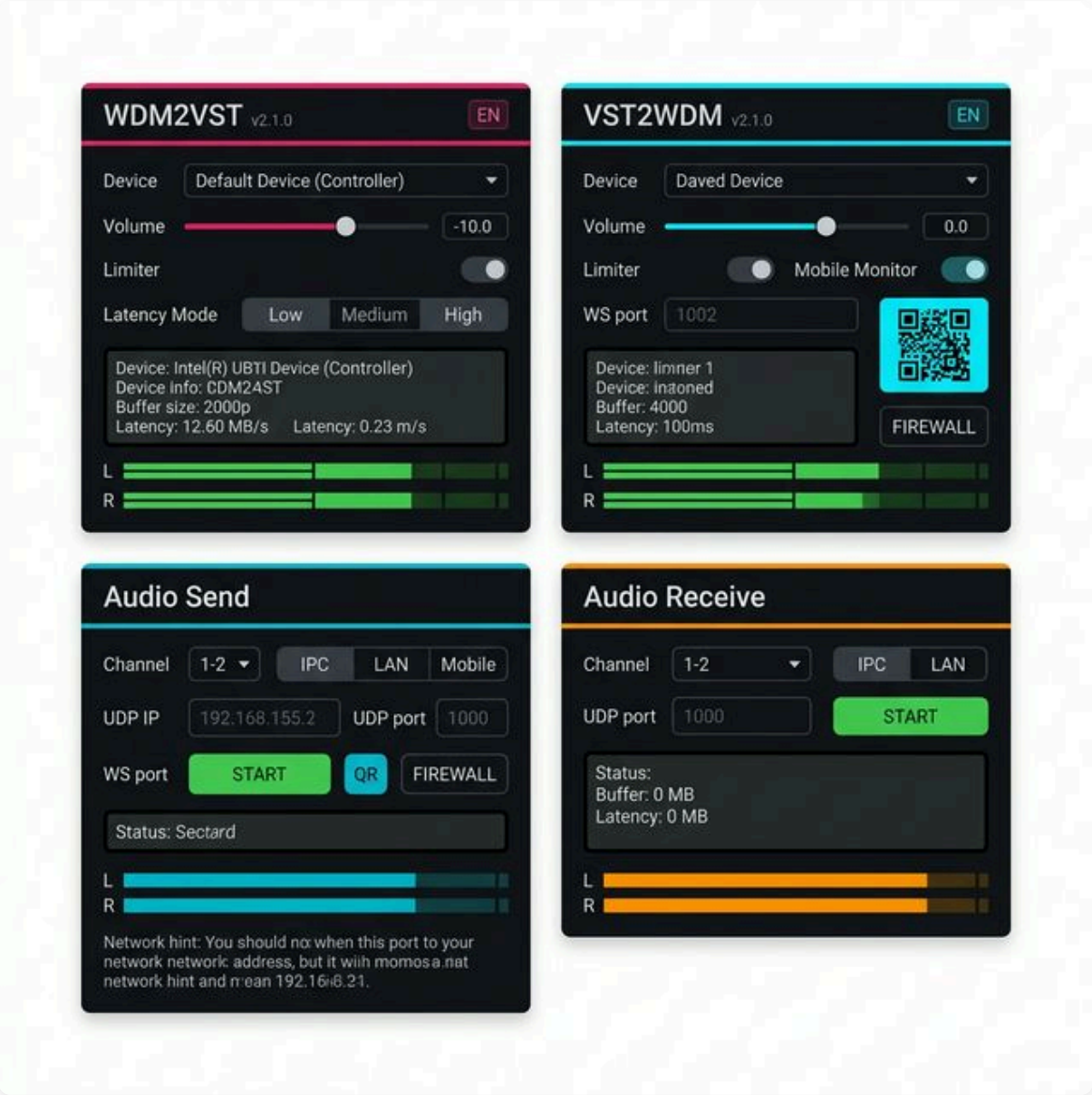


Figure 2: Four VST3 Plugin Interfaces

## 4.1 WDM2VST — System Audio → DAW

**Purpose:** Capture any system audio (music, browser, game, etc.) into a DAW track.

### Controls

| Control | Description |
|---|---|
| **DEVICE** | Select audio input source |
| **LEVEL** | Adjust input volume (0 – 200%) |
| **Limiter** | Soft limiter to prevent clipping |
| **Latency Mode** | Safe (8ms+) / Fast (4ms+) / Extreme (2ms+) / Insane (1ms+) |
| **CN/EN** | Chinese / English language toggle |

### Connection Modes

| Mode | Selection | Latency |
|---|---|---|
| Standard Recording | Any microphone/line input | Hardware dependent |
| System Loopback | Loopback / Speaker device | ~20ms |
| **IPC Channel** | Virtual 1/2 [IPC] etc. | **~0ms (zero latency)** |

> **Tip:** IPC mode provides zero-latency routing. Select the IPC channel matching the Virtual Speaker endpoint used by your system audio.

### Quick Start: Record System Audio

1. Set Windows default playback device to **Virtual 1/2**
2. Insert **WDM2VST** on a DAW track
3. Select **Virtual 1/2 [IPC]** from the device dropdown
4. Play audio on your PC — it appears on the DAW track

## 4.2 VST2WDM — DAW → System Audio + Mobile

**Purpose:** Route DAW audio to physical speakers/headphones and optionally to phone monitoring.

### Controls

| Control | Description |
|---|---|
| **DEVICE** | Select audio output device |

| | |
|---|---|
| **LEVEL** | Output volume (0 – 200%) |
| **Limiter** | Soft limiter |
| **Mobile** | Enable phone monitoring via WebSocket |
| **WS Port** | WebSocket port (default: 18820) |
| **QR** | Show QR code for phone connection |
| **Fix FW** | Auto-configure Windows Firewall |

### Mobile Monitoring Steps

1. Enable the **Mobile** toggle
2. Click **Fix FW** (first time only)
3. Click **QR** — scan with your phone camera
4. Audio plays in phone browser in real-time

> **Note:** Phone and PC must be on the same WiFi network.

## 4.3 Audio Send — DAW → Network

**Purpose:** Stream DAW audio to other PCs or mobile devices.

| Control | Description |
|---|---|
| **CHANNEL** | P2P channel 1–8 (must match receiver) |
| **IPC/LAN/Mobile** | Enable transport modes |
| **LAN TARGET** | 127.0.0.1 (same PC) / 192.168.x.x (other PC) / .255 (broadcast) |
| **UDP/WS PORT** | Network ports |
| **START/STOP** | Start or stop sending |

## 4.4 Audio Receive — Network → DAW

**Purpose:** Receive audio from Audio Send on this or another PC.

| Control | Description |
|---|---|

| | |
|---|---|
| **CHANNEL** | Must match sender’s channel |
| **IPC/LAN** | Enable transport modes |
| **UDP PORT** | Must match sender’s port |
| **START/STOP** | Start or stop receiving |

# 5. Common Scenarios

## 5.1 Record System Audio

```
[Windows Audio] → Virtual 1/2 → IPC → WDM2VST → DAW Track
```

## 5.2 Monitor DAW on Phone

```
[DAW Master] → VST2WDM (Mobile ON) → WebSocket → Phone Browser
```

## 5.3 Stream to Another PC

```
[DAW-A] → Audio Send (LAN) → UDP → Audio Receive → [DAW-B]
```

# 6. Troubleshooting

| Problem | Solution |
|---|---|
| No virtual audio devices | Reinstall driver, check Device Manager |
| IPC channel no audio | Verify plugin and driver use same channel ID |
| Mobile QR scan fails | Click Fix FW, check same WiFi network |
| LAN no audio received | Check firewall, verify IP and port match |
| Audio crackling | Increase DAW buffer size or use Safe latency mode |

# 7. Language Support

All plugins support **Chinese / English** toggle via the **CN/EN** button (top-right corner). Default: Chinese.